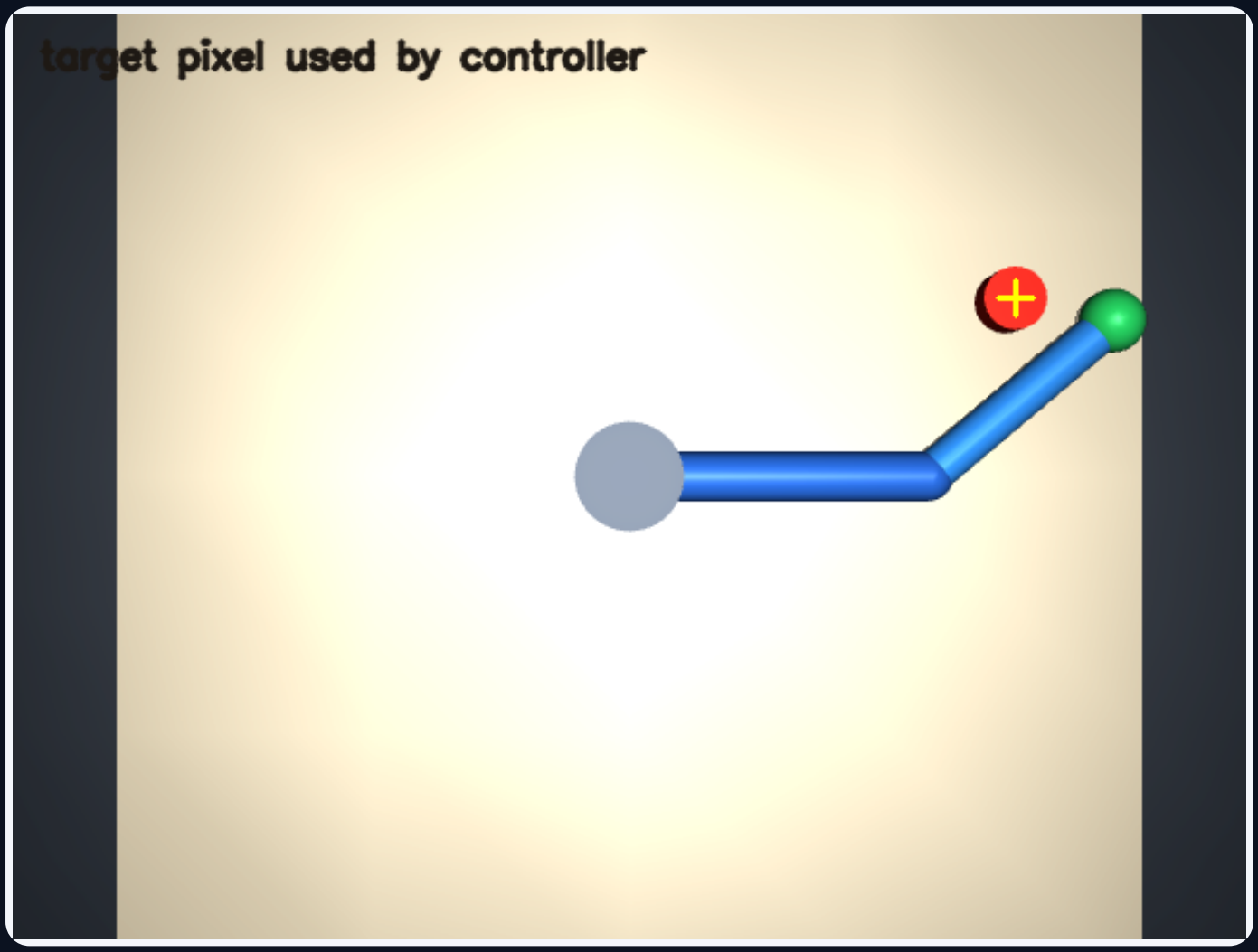
ROBOTICS PORTFOLIO | PROJECT GUIDE

Camera-Based Visual Servoing

From planar servoing to multi-arm contact



Pixels -> a table-plane estimate -> joint motion -> a measured result

This guide explains the important code, what the benchmarks actually establish, which experiments were rejected, and the next evidence-based milestone.

MuJoCo | OpenCV | Image Jacobian | Paired benchmarks

Local work on feat/cooperative-multiarm-pick-demo | Updated 22 Oct 2026

The system in one picture

A small simulation that makes every perception and control step inspectable.

01

Render

MuJoCo produces a top-down RGB image of the table, target, and arm.

02

Detect

OpenCV segments the red target and green end-effector marker.

03

Estimate

A pinhole camera model maps target pixels to table coordinates.

04

Move

Inverse kinematics or image feedback updates the two arm joints.

05

Score

MuJoCo ground truth measures final distance after control stops.

What is measured and what is not

The controller receives rendered pixels and joint state. Ground-truth target position is read after the run for scoring; it is not fed into the controller.

What one feedback update does

Image feedback corrects pixel error with a local model of how each joint moves the image.

DAMPED IMAGE-SPACE UPDATE

$$e = target_pixel - tip_pixel$$

$$dq = 0.5\ J^T (J\ J^T + I)^{-1}\ e$$

then clip each joint change to 0.30 rad and advance 0.4 simulated seconds

1. Measure

Find red target and green tip pixels.

2. Compare

Compute the 2D image-space error.

3. Solve

Use the Jacobian and damping term to get dq.

4. Advance

Move, render again, and count a fresh check.

Why the damping term is there

The arm has two joints but the observed error has two pixel coordinates. Near a singular pose, a raw inverse can demand a huge joint change. Adding I stabilizes the solve, and the per-joint cap limits one update.

Two different tip sources in experiments

The standard image-feedback controller detects the tip in the camera image. The timing benchmark can instead project the current joint angles through the calibrated geometry. That is current proprioceptive state paired with a delayed target image, not fully delayed vision.

Important constants in run.py

8 px tolerance | 3 consecutive checks | 20 maximum motion updates | 5 missing-target observations of grace | 4.5 cm physical success radius

The controller stop reason and the physical reach score are separate fields in each trial.

Where the important code lives

The core loop is intentionally small. Benchmarks isolate one assumption at a time.

model.py — MODEL_XML
Inline scene: camera, table, two-link arm, target, and position actuators.

controller.py — projection + IK + Jacobian
Camera geometry, forward/inverse kinematics, pixel Jacobian, damped joint update, and opt-in predictor functions.

run.py — run_trials
Detection, seeded targets, control loop, dropouts, camera delay, visibility trace, stop reason, physical scoring, and demo recording.

benchmark.py — paired robustness suite
Same seeded targets across open loop and image feedback; aggregates success and error percentiles.

dynamic_benchmark.py — moving target + latency
Separates target motion, observation age, and strict vs experimental stopping.

prediction_benchmark.py / reversal_benchmark.py — opt-in predictor tests
Compares measured target with extrapolators while keeping current joint-state tip projection explicit.

sensor timing + timestamped verification — feedback + completion
Compares fresh, fully delayed, and hybrid control inputs; tests post-stop camera age and settling.

motion_envelope_benchmark.py / conditional_uncertainty_benchmark.py — held-out completion
Calibrates global or pixel-feature q99 margins on training seeds, then scores fresh visual decisions on held-out camera replays.

verification_*.py / terminal_verification.py — fresh-view evidence + quality
Renders a fresh terminal view, scores marker quality, and measures calibration sensitivity, coverage, and decision errors.

camera_calibration_*.py — landmark health + transfer
Matches fixed landmarks to pinhole projections, freezes a threshold, then evaluates pose transfer, visibility, and radial distortion.

public_camera_calibration.py / camera_distortion.py — fit and correct radial lens error
Fits k1,k2 on recorded checkerboard views; applies optional pixel correction and benchmarks its effect on reaching.

perception_candidate.py / perception_candidate_validation.py — circle prior + held-out test
Keeps the largest-contour baseline and compares an opt-in size-prior detector with abstention on randomized rendered scenes.

Good first files to read

Start with model.py, then controller.py, then run.py around run_trials and the image_feedback branch. Read benchmark modules after you can trace one episode.

What the current controller handles

Image feedback corrects several errors, but noisy sensing and visibility still expose real failure modes.

PHYSICAL SUCCESS BY TARGET-PIXEL NOISE



Blue: open loop Green: image feedback | 100 trials per condition

At 20 px noise

The default feedback controller reached 88/100 targets within 4.5 cm; open loop reached 34/100. Feedback median error was 2.12 cm and P95 was 6.39 cm. The controller uses a three-sample median, but this does not eliminate large-noise failures.

DROPOUT LENGTH SEPARATES RECOVERY FROM TIMEOUT

Injected blackout	Physical success	Reacquired after gap
1 observation	100/100	100/100
3 observations	100/100	100/100
5 observations	100/100	100/100
6 observations	33/100	0/100

Ground-truth distance decides physical success. The controller can stop without fresh vision even when the arm is physically close.

Natural self-occlusion is the next measured gap

In the clean strict run, 91/100 trials reached physically but stopped after the arm hid the target from the overhead camera.

Three sensor setups that must stay distinct

The timing experiments separate image age from the source used to locate the arm tip.

Fresh RGB

target pixel now | tip pixel now

Fully delayed RGB

target pixel delayed | tip pixel delayed

Vision + proprioception

target pixel delayed | tip from current joints

How to read the held-out results

Each row uses seeds 100-104 and 100 trials. The delta is paired median reaching error: hybrid minus fully delayed RGB. A negative value means lower error in this simulator.

Target path	Noise	Delay	Delayed RGB	Hybrid tip
Sinusoidal	0 px	0.4 s	4%	84%
Sinusoidal	0 px	0.8 s	63%	81%
Sinusoidal	2 px	0.4 s	11%	68%
Sinusoidal	2 px	0.8 s	56%	85%
Piecewise linear	0 px	0.4 s	34%	88%
Piecewise linear	0 px	0.8 s	58%	93%
Piecewise linear	2 px	0.4 s	37%	83%
Piecewise linear	2 px	0.8 s	56%	95%

What improved, and what did not

The current-joint-state tip hybrid improved physical success in all eight held-out settings. Its P95 error was 4.92-6.71 cm. It still had 0/100 current-time visual confirmations in every delayed condition. Physical reaching and fresh visual completion remain separate outcomes.

The paired bootstrap resamples only five seed groups. The interval reflects these simulator runs, not broad generalization.

A stopped arm needs time-correct visual evidence

The controller's reaching result and a camera's later completion report are separate events. This benchmark replays the stopped arm while a delayed camera continues observing a moving target.

Controller stops

Save measured joint pose and velocity; hold the pose.

Arm settles

Joint speed <0.02 rad/s for 0.1 s; within 0.8 s.

Camera captures

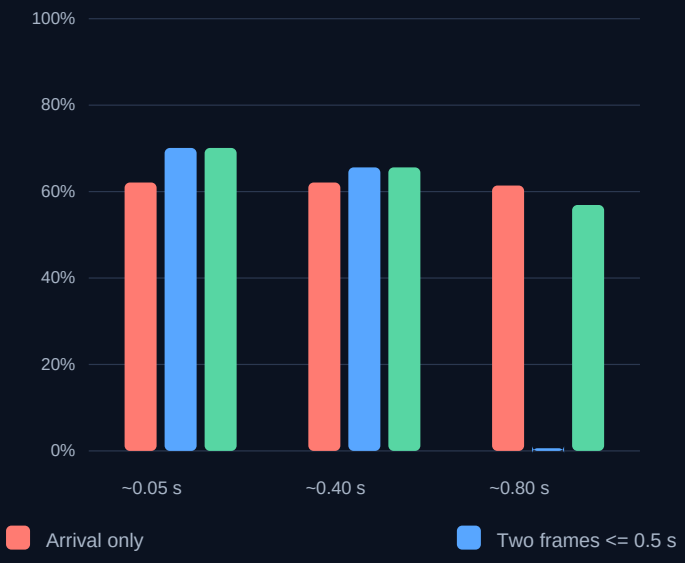
5 Hz stream; require two consecutive post-settle frames.

Receiver decides

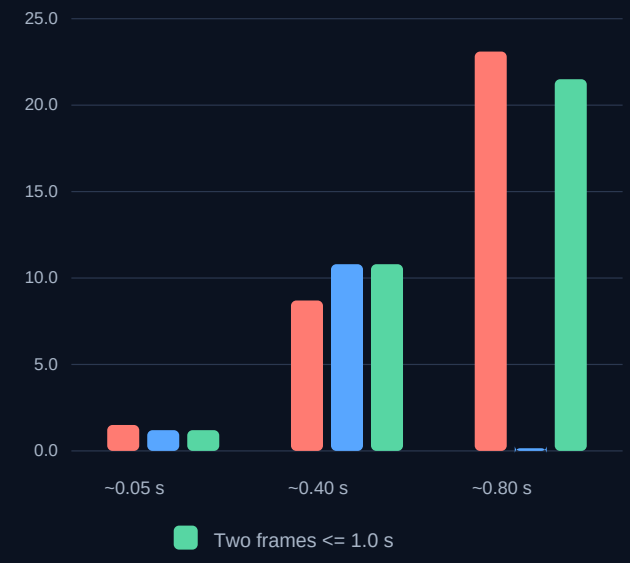
Check frame age and deliver by a 2 s deadline.

OUTCOME RATES ACROSS ALL MOTION PROFILES AND CONTROL MODES

Visual confirmation at delivery



Outside 4.5 cm at delivery per 100 trials



1,000 controller trials | 9,000 policy outcomes | seeds 900-904 | four start phases per motion path

The age limit changes coverage and delivery-time reliability

Across 3,000 0.5 s guarded outcomes, 120 image-positive decisions arrived after the target-arm distance left the success radius; 108 were physically within the radius at capture. At 1.0 s, 335 mismatches occurred, and 318 were within the radius at capture. Most mismatches therefore reflect target motion during transport, not an incorrect image at capture. These age limits remain measured tradeoffs, not validated defaults.

Keep the two outcomes and the three clocks explicit

Physical reaching is scored at controller stop from MuJoCo state. Visual confirmation uses camera estimates from frames captured after the arm settles. Delivery-time distance is an evaluation label only; it is never sent to the detector, controller, or confirmation rule. This is an offline benchmark and does not change the default controller. The next investigation should test conservative task-completion semantics across target speeds, frame ages, and transport loss before any runtime policy is promoted.

Target motion makes stale camera decisions visible

The follow-on benchmark holds the stop, settling, camera, and confirmation pipeline fixed while varying target path and analytic peak speed.

Most stale positives began during transport

For the 0.5 s age guard, 84/90 positive decisions were physically inside the radius at image capture. For the 1.0 s guard, 219/225 were close at capture. Every stationary-target guard cell had zero delivery mismatches.

TWO-FRAME POLICY: CONFIRMATION AND STALE-AT-DELIVERY RATES

	Maximum frame age 0.5 s			Maximum frame age 1.0 s		
	0.05 s	0.40 s	0.80 s	0.05 s	0.40 s	0.80 s
Stationary 0.000 m/s	confirm 92.0% out 0.0% (0/75)	confirm 92.0% out 0.0% (0/75)	confirm 0.0% out 0.0% (0/75)	confirm 92.0% out 0.0% (0/75)	confirm 92.0% out 0.0% (0/75)	confirm 92.0% out 0.0% (0/75)
Sine 0.25 Hz 0.063 m/s	confirm 60.7% out 0.3% (1/300)	confirm 59.0% out 9.0% (27/300)	confirm 0.0% out 0.0% (0/300)	confirm 60.7% out 0.3% (1/300)	confirm 59.0% out 9.0% (27/300)	confirm 52.7% out 18.0% (54/300)
Sine 0.50 Hz 0.126 m/s	confirm 95.0% out 2.0% (6/300)	confirm 91.7% out 10.7% (32/300)	confirm 0.0% out 0.0% (0/300)	confirm 95.0% out 2.0% (6/300)	confirm 91.7% out 10.7% (32/300)	confirm 84.0% out 17.3% (52/300)
Piecewise 0.50 Hz 0.080 m/s	confirm 96.0% out 2.3% (7/300)	confirm 94.0% out 5.7% (17/300)	confirm 0.0% out 0.0% (0/300)	confirm 96.0% out 2.3% (7/300)	confirm 94.0% out 5.7% (17/300)	confirm 89.3% out 9.7% (29/300)

A fixed age cap is not a general completion rule

The 0.5 s cap confirmed no high-latency outcomes, while 1.0 s restored coverage and allowed motion-related mismatches. Reliability was not monotonic in peak speed: the 0.25 Hz sinusoid had more stale delivery positives than either fast path. Path shape and starting phase also matter.

Keep reaching separate from visual completion

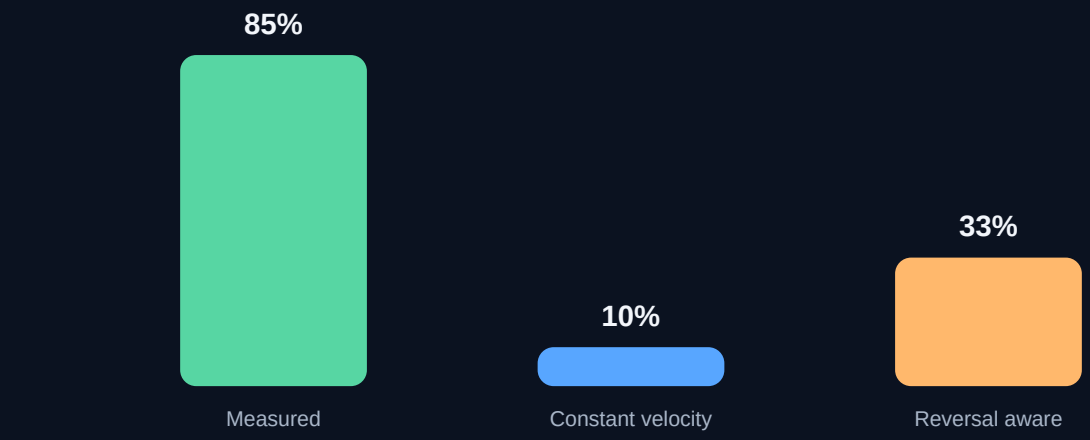
Controller-stop physical reaches: Stationary 74/75; Sine 0.25 Hz 176/300; Sine 0.50 Hz 200/300; Piecewise 0.50 Hz 228/300. The rate heatmap instead counts fresh camera decisions. The five controller modes remain distinct in CSV; only this chart pools them within each condition and delivery profile. Delivery-time ground truth is evaluation-only.

The 975 controller trials use seeds 950–954 and produce 8,775 policy outcomes. Camera motion, pixel noise, frame loss, settling, and the controller are unchanged from the previous timestamped benchmark. Results cover one target amplitude and five seed groups; no real camera or hardware was tested.

A useful negative result: prediction lost to measurement

Moving-target latency made extrapolation tempting. Paired evidence says these predictors stay experimental.

SINUSOIDAL TARGET | 0.4 S DELAY | NOISE-FREE



Success within 4.5 cm | five seeds x 20 targets

Method	Median reaching error	Success
Measured target	2.70 cm	85%
Constant velocity	6.48 cm	10%
Reversal aware	6.44 cm	33%

Across all eight reversal benchmark settings

The reversal-aware predictor increased median error in every setting; each seed-cluster bootstrap interval for the paired error delta was above zero. It often beat plain velocity prediction, but never beat the measured-target baseline on success. Keep predictors opt-in and preserve the baseline.

At 0.8 s delay the reversal-aware method reached only 5% on sinusoidal noise-free trials versus 85% for measured target. The measured result rules out this simple reversal guard as the default latency fix.

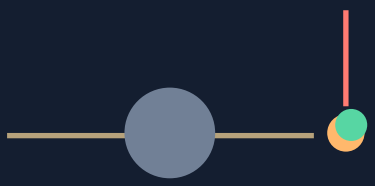
Other measured tradeoffs

The three-sample median helped 20 px noise versus raw feedback, while larger windows raised clean error. Stale-target stopping did not improve physical success and can stop without reacquisition. These remain scoped experiments, not new defaults.

A second view can supply a separate visual estimate

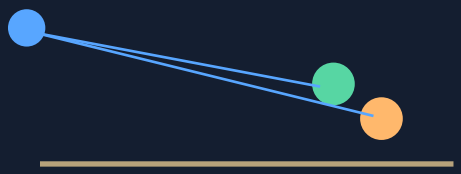
Four oblique candidates were evaluated; an opt-in east-view check now runs after the overhead controller stops.

OVERHEAD CONTROL VIEW



The arm can cover the target from above.

EAST OBLIQUE CANDIDATE



Pixels are projected to known marker heights.

View	Control noise	Both seen	False success	Missed success
East	0 px	200/200	0	1
East	20 px	198/200	0	14
North	0 px	200/200	0	0
North	20 px	200/200	6	11

Why east is the current candidate

Across the two seed groups it produced no false success estimates and saw both markers in 198/200 noisy trials. It missed 14 true noisy-condition successes. The north view saw every marker but made six false success estimates. The conservative tradeoff is promising, not sufficient for a production stop rule.

Scope of this evidence

The oblique camera is rendered from the simulator. The controller still uses the overhead camera, and terminal verification remains a separate fresh observation after control stops. The next page measures estimator calibration offsets and controlled image changes.

The opt-in --verify-terminal path now renders the east camera after control stops. It reports the fresh image estimate separately; it never changes the controller stop or MuJoCo physical score. Missing markers make the estimate unavailable.

Calibration error changes the visual decision

100 held-out paired trials per condition; each verifier stress leaves the reaching controller unchanged.

100
paired trials per condition

0 / 8
controller outcome changes

85 / 100
physical reaches in every row

Verifier condition	Both seen / estimate	False +	Missed	P95 error
Nominal	99 / 100	0	7	1.74 cm
Estimated FOV +5°	99 / 100	0	15	2.12 cm
Estimated FOV -5°	99 / 100	1	3	1.66 cm
Camera world X +2 cm	99 / 100	0	7	1.74 cm
Estimated yaw +3°	99 / 100	0	5	1.84 cm
Exposure ×0.25	0 / 100	n/a	n/a	n/a
Exposure ×1.5	99 / 100	0	10	2.52 cm
Target occluded	0 / 100	n/a	n/a	n/a
Tip occluded	0 / 100	n/a	n/a	n/a

MISSED PHYSICAL SUCCESSES AMONG AVAILABLE ESTIMATES



What these stresses do and do not establish

A +5° FOV estimate doubled missed successes (7 to 15); -5° caused one false success. Dim exposure and marker occlusion yielded no estimate, so those trials have no decision score. Brightness is a uniform post-render multiplier and occlusion uses marker-color removal plus inpainting. This is a simulator stress test, not real-camera evidence.

The quality gate trades coverage for fewer errors

A heuristic marker score was swept on the same held-out trials; these results do not select a deployment threshold.

01 **Contour area**

02 **Circularity**

03 **Saturation**

04 **Brightness**

Score = minimum component across both detected markers; it is a heuristic, not a probability.

Minimum score	Coverage	False +	Missed	Condition
0.00	99 / 100	1	3	FOV estimate -5°
0.15	93 / 100	1	3	FOV estimate -5°
0.20	39 / 100	1	3	FOV estimate -5°
0.50	9 / 100	0	1	FOV estimate -5°

AVAILABLE ESTIMATES AFTER APPLYING THE GATE



A useful abstention point has not been found

The 0.50 gate removed the single false success under a -5° FOV estimate, while leaving 9% coverage. At 0.20, coverage was 39% and the false success remained. Scores do not reveal calibration bias from marker appearance; no threshold is selected.

The frozen quality gate failed its coverage target

Threshold chosen on seeds 200-204, then evaluated unchanged on seeds 300-304; each split has 100 target trials paired across nine image conditions.

Calibration rule

Choose maximum candidate coverage with zero observed false-success decisions; require at least 50% coverage for a usable gate.

INDEPENDENT VALIDATION RESULTS

Gate	Accepted / candidate	Coverage	False +	Missed
Ungated 0.00	600 / 600	100%	10	48
Frozen 0.60	23 / 600	3.8%	0	3

CANDIDATE-ESTIMATE COVERAGE



The gate acted as a brightness filter

All 23 accepted estimates came from bright exposure. It accepted none from nominal images or any camera calibration shift. Across all 900 validation condition-trials, 23 estimates were available (2.6%); dim and occluded images had no candidate detection before gating.

Paired held-out video

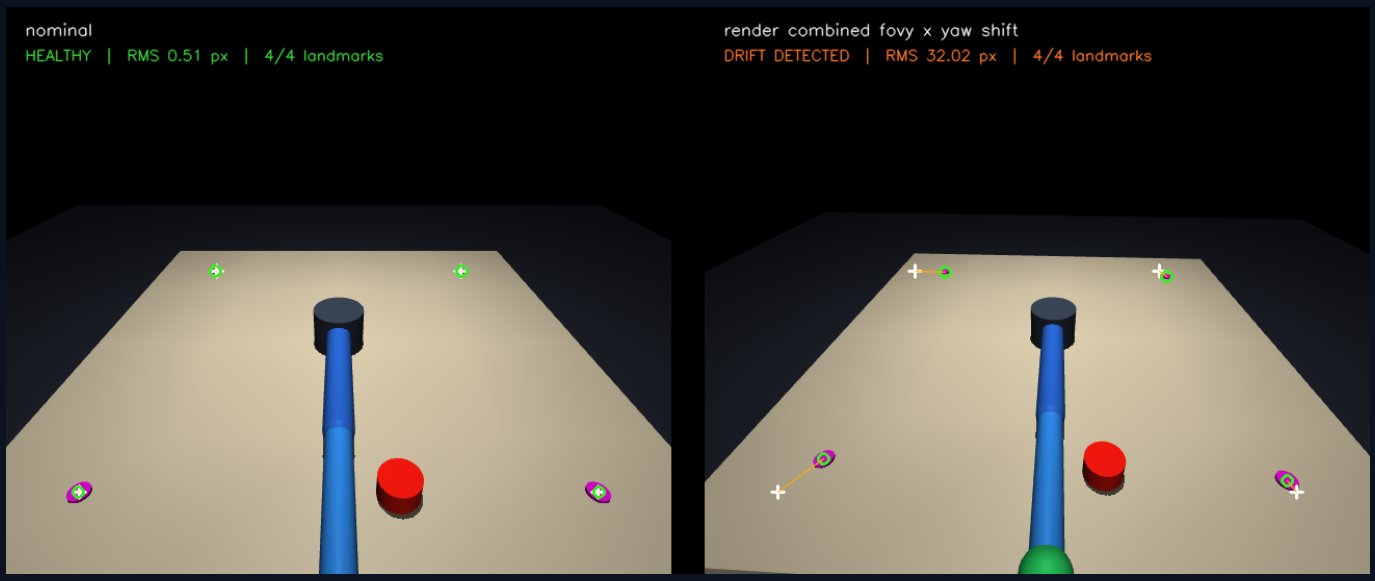
Seed 302, episode 17: fresh-view estimate 4.494 cm, MuJoCo physical error 4.6 cm. The 0.60 gate abstains; both clips preserve the same reaching outcome.

Negative result: no usable threshold found

The gate removed 10 false visual success decisions on validation but retained only 3.8% of candidate estimates, below the 50% usability floor. Zero observed errors do not establish a risk guarantee; no threshold is recommended.

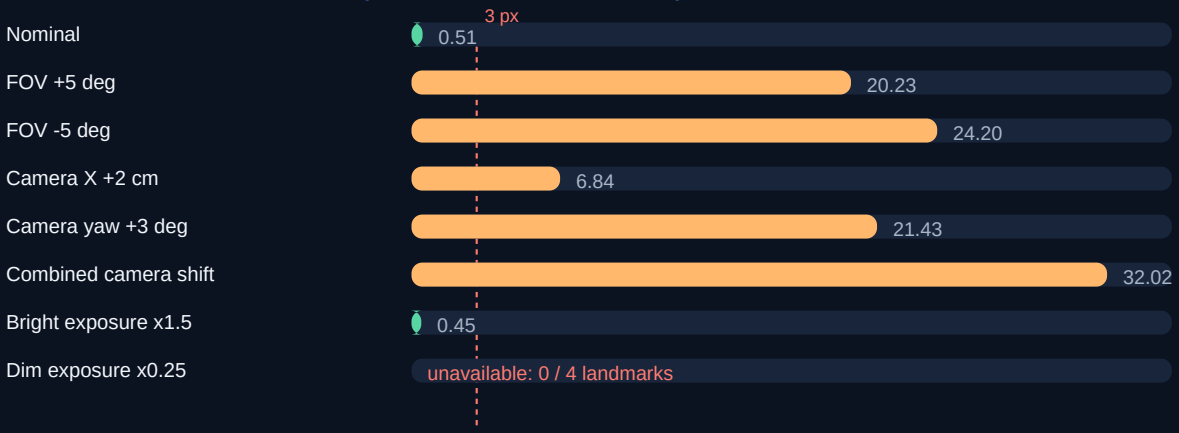
Independent landmarks expose camera-model drift

A separate simulator diagnostic compares known tabletop points with image detections while the rendered camera moves away from nominal calibration.



White crosses: nominal projection | Green rings: detected centers | Orange lines: residuals

MEASURED REPROJECTION RMS (PIXELS; ALARM LINE AT 3 PX)



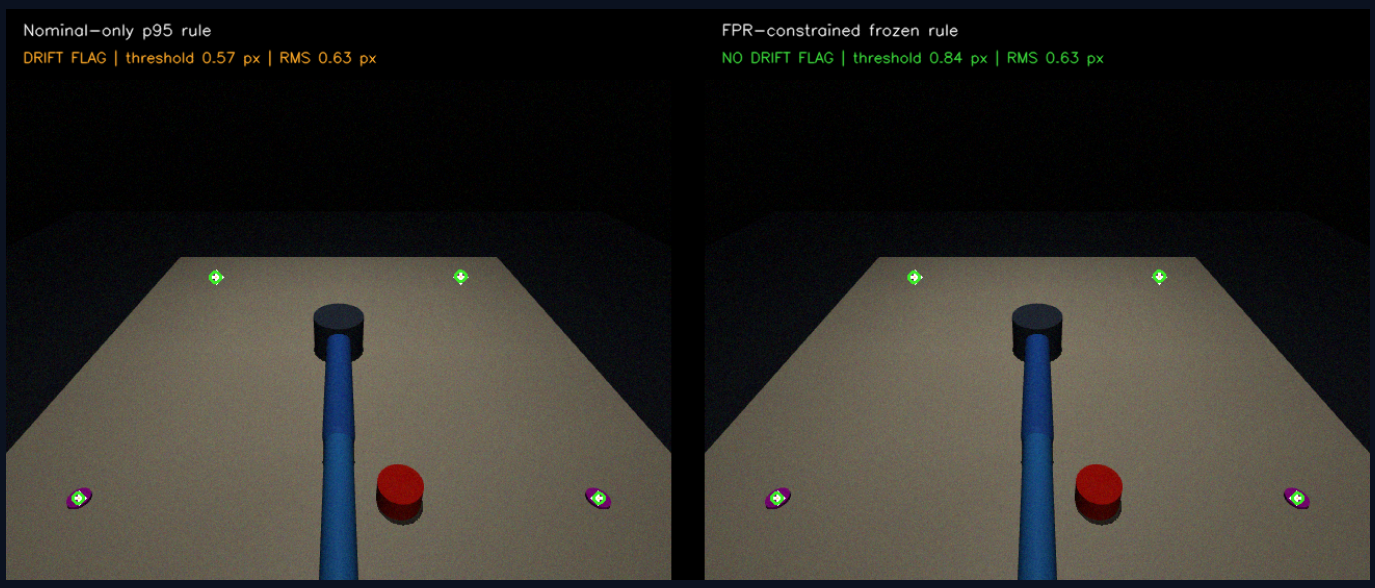
01 Known table coordinates → **02** Nominal pinhole projection → **03** Magenta center detections → **04** Match + pixel RMS residual

Promising detectability; threshold still exploratory

Five renderer-side camera shifts exceeded 3 px RMS. Nominal and bright-exposure images remained below it; dim images were unavailable. The 3 px threshold has not been validated on independent or real-camera frames. This diagnostic neither updates calibration nor changes reaching.

Labelled calibration shifts improved the threshold

A threshold was selected on seeds 400-404, frozen, then evaluated on fresh seeds 500-504 with broader image noise, brightness, and blur.



The same held-out nominal frame triggers the lower threshold but not the frozen supervised threshold.

HELD-OUT RESULTS (AVAILABLE LANDMARK ESTIMATES)

Threshold policy	RMS px	Nominal false alarms	Subtle	Moderate	Severe
Nominal-only p95	0.573	7 / 95 (7.4%)	100 / 100	100 / 100	100 / 100
Supervised, 5% cap	0.841	0 / 95	100 / 100	100 / 100	100 / 100
Fixed exploratory	3.000	0 / 95	60 / 100	93 / 100	100 / 100

CALIBRATION RULE

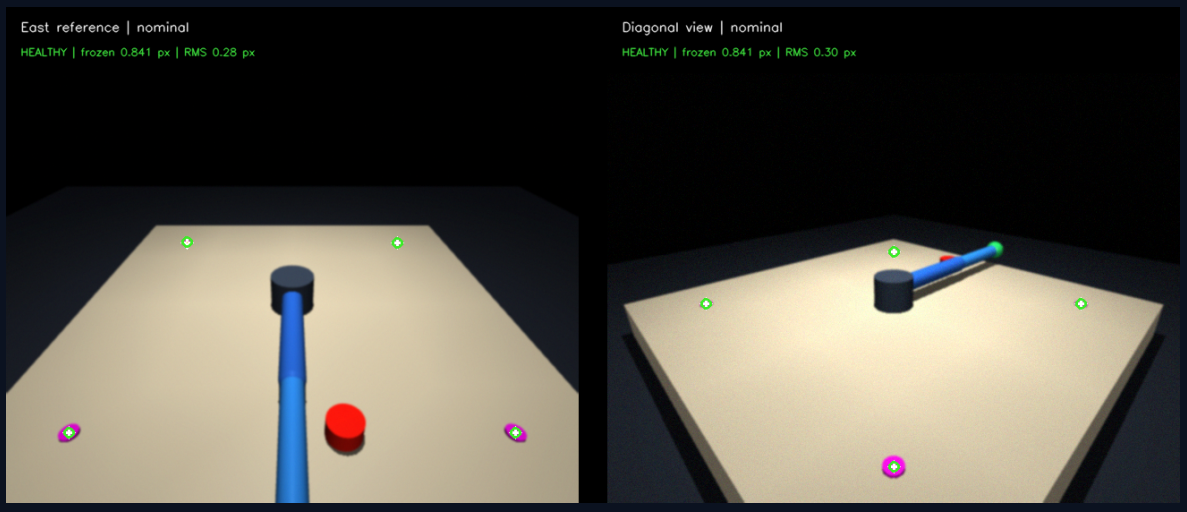
The supervised rule maximized shifted-camera detection on labelled calibration renders while keeping nominal calibration false alarms at or below 5%. It selected 0.841 px, with 0/100 nominal false alarms and 300/300 calibration shifts detected.

Promising result with a narrow validation boundary

On held-out seeds, the supervised rule flagged 0/95 nominal images with visible landmarks and detected all 300 new camera shifts. Five dim probes were unavailable. The study reuses one scene, landmark layout, shift families, and severity ranges; it is not a real-camera or reaching-success result.

Camera pose changes the residual evidence

The 0.841 px East-view threshold was frozen and tested on 3,000 simulator renders across five camera poses and synthetic radial lens warps.



White crosses: expected centers | Green rings: detections | Orange lines: candidate residuals

FROZEN-THRESHOLD CAMERA-SHIFT RESULTS

Pose	Nominal alarms	Subtle	Moderate	Severe / unavailable
East reference	0/95	100/100	100/100	100/100; 0 unavailable
West lower	2/95	100/100	99/99	82/82; 18 unavailable
North higher	0/95	99/100	100/100	100/100; 0 unavailable
Diagonal	0/95	100/100	100/100	99/99; 1 unavailable
East grazing	Unavailable	Unavailable	Unavailable	Unavailable

RADIAL DISTORTION ALARMS (25 IMAGES PER COEFFICIENT)

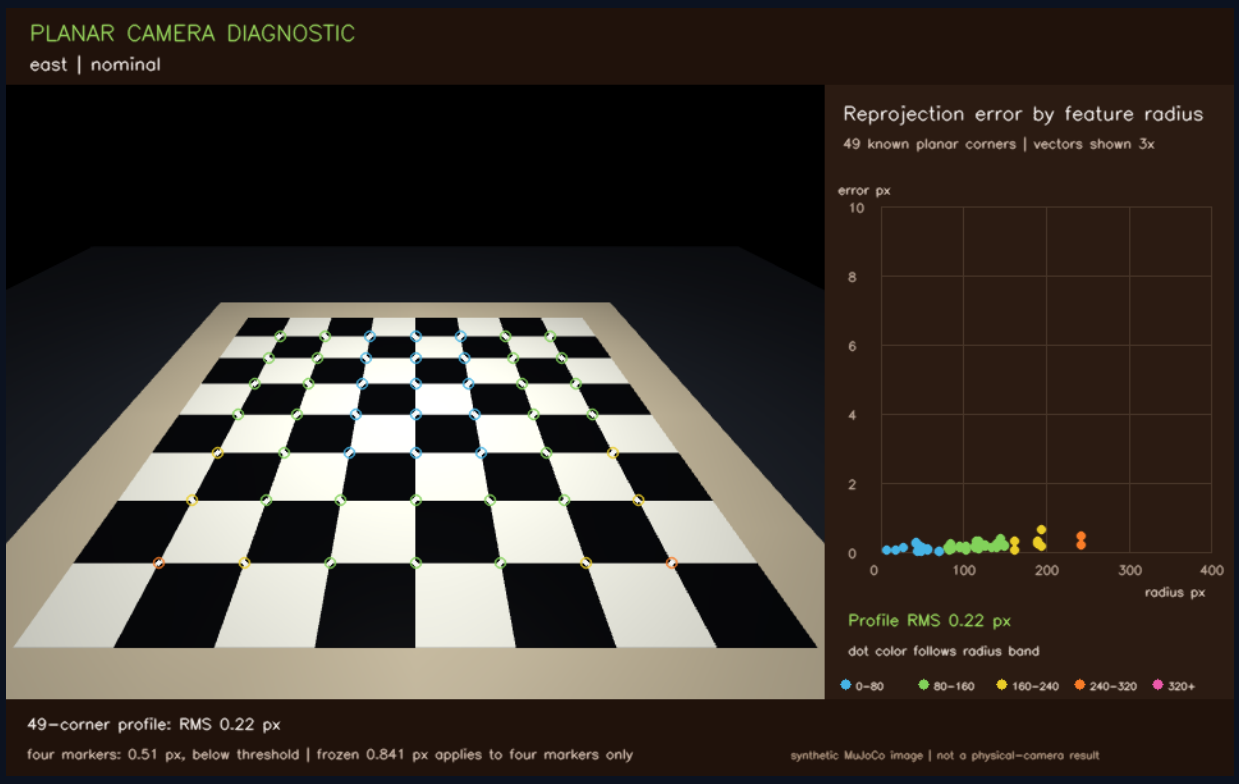
Synthetic k1	East	North higher	Southwest diagonal
-0.005	0/25	0/25	0/25
+0.005	25/25	0/25	0/25
+0.010	25/25	0/25	1/25
+0.020	25/25	25/25	25/25

A frozen pixel threshold does not transfer uniformly

The same +0.010 radial warp triggered 25/25 East images but 0/25 higher-North images. Edge-clipped expected points produce an unavailable result; their raw candidate RMS is retained. A dense planar follow-up now profiles residuals by radius. Next: validate it on recorded camera frames.

Residuals now have a radius

A 49-corner planar target adds spatial coverage to the frozen four-marker camera check. The profile is descriptive; the 0.841 px threshold remains specific to the original markers.



Color marks expected image radius; white dots show detected corners; residual vectors are exaggerated 3x.

NOMINAL SIMULATOR COVERAGE (FIVE APPEARANCE SAMPLES PER VIEW)

Pose	Grid available	Grid max radius	4-marker max radius	Grid median RMS
East	5/5	241 px	297 px	0.28 px
West lower	5/5	272 px	343 px	0.79 px
North higher	5/5	186 px	223 px	0.16 px
Southwest diagonal	4/5	182 px	210 px	0.41 px
East grazing	3/5	287 px	366 px	1.59 px

Radius explains a view-dependent miss

East k1=+0.010: median signed radial residual is -0.05 px at 0-80 px and +0.86 px at 240-320 px. The frozen four-marker rule alarms on 5/5 East images, then 0/5 higher-North images at the same synthetic coefficient.

More field coverage costs visibility

A 0.17 m-square probe reached a 354 px radius but fell to 2/5 West, 4/5 diagonal, and 0/5 grazing availability. The 0.15 m-square target keeps higher availability while undersampling some edge radii. All evidence remains synthetic.

Distortion correction is conditional

A public-camera fit checks the calibration pipeline; a paired MuJoCo study checks whether optional pixel correction changes reaching.

RECORDED-CAMERA CALIBRATION: ONE OPENCV SEQUENCE

8 training views
fit focal lengths and radial k1,k2

→

Freeze intrinsics
estimate each held-out board pose

→

4 held-out views
measure corner reprojection error



Held-out corners stop within 240 px of the estimated image center. The camera has no published ground-truth intrinsics, so this is a one-sequence fit check, not proof of a better robot camera model.

MUJOCO REACHING: NO CORRECTION VERSUS EXACT INJECTED K1



20 PX NOISE: PAIRED SUCCESS CHANGES AND ERROR TAILS

Controller	k1	No correction success; med/P95	Exact k1 success; med/P95	Recovered / lost
Open loop	-0.10	28/100; 6.46/13.98 cm	29/100; 6.65/13.94 cm	3 / 2
Open loop	+0.10	31/100; 6.39/13.81 cm	33/100; 6.39/13.86 cm	2 / 0
Image feedback	-0.10	81/100; 2.67/6.18 cm	79/100; 2.78/6.26 cm	6 / 8
Image feedback	+0.10	83/100; 2.75/6.35 cm	82/100; 2.29/6.86 cm	7 / 8

Correction did not fix the high-noise failure

At 20 px, recovered and lost success counts were close; feedback lost one or two successes. Positive k1 feedback improved its median but worsened P95 by 0.51 cm. The 2 px chart above has 100/100 successes in every row. Exact coefficients are an oracle check, +/-0.10 is a simulation stress value, and fresh visual confirmation was not evaluated.

A strict envelope cuts stale positives, with low coverage

A training-calibrated image margin and target-speed cap reduce delivery mismatches, but this held-out result does not justify a runtime rule.

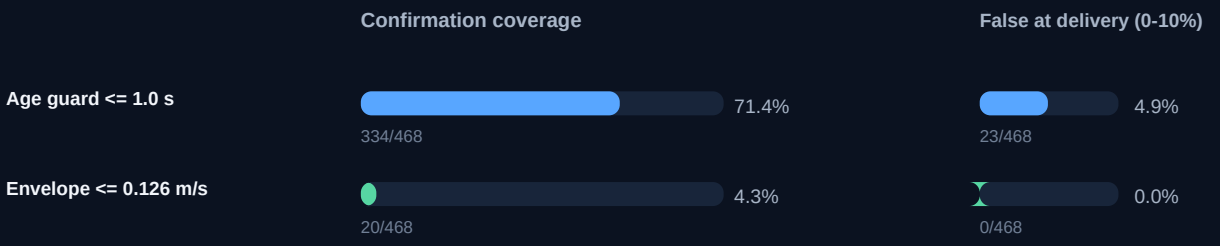
2.54 cm
empirical q99 margin

8.2%
envelope coverage

0 / 2,340
false confirmations

23 / 16,168
delivery-event misses

FULLY DELAYED RGB | TWO-OBSERVATION CONTROL DELAY



The empirical margin is not a guarantee

Across the three profile-specific delivery streams, the q99 image margin underbounded capture distance on 202/16,168 events and the motion envelope exceeded delivery distance on 23/16,168. The streams replay the same captures with different loss and age, so records are correlated. Two-frame confirmation made no false decisions here, but event-level violations remain.

Control input	Age guard: confirms; false	Envelope: confirms; false	Physical reach at stop
Fresh RGB	412/468; 33	49/468; 0	137/156
Fully delayed RGB, 1 obs	387/468; 52	25/468; 0	91/156
Fully delayed RGB, 2 obs	334/468; 23	20/468; 0	90/156
Delayed target + current tip, 1 obs	413/468; 38	40/468; 0	117/156
Delayed target + current tip, 2 obs	403/468; 39	58/468; 0	125/156

Each camera-policy cell pools three delivery profiles (n=468). Physical reaching is scored at controller stop over 156 trials per mode; the later camera decision remains a separate result.

Why calibrate image uncertainty conditionally?

A single 2.54 cm q99 margin reduced held-out coverage to 8.2%. Pixel location changes how camera rays map to the table, while marker-localization quality varies across frames. The next page reports the held-out test of those image-only signals. Completion policies stay offline.

REPRODUCE THE BENCHMARK AND PAIRED DEMO

```
uv run visual-servo-motion-envelope-benchmark

uv run --extra demo python scripts/generate_motion_envelope_demo.py
```

No runtime controller change; image confirmation and controller-stop physical reaching remain separate.

The held-out comparison keeps the decision evidence separate

A three-band empirical q99 uses only pixel-derived uncertainty features at decision time; simulator distances score coverage after the fact.

2.59 cm
new global q99

3
conditional feature bands

10,914
training camera frames

4
held-out seeds

FULLY DELAYED RGB | TWO-OBSERVATION CONTROL DELAY

Policy	Confirmation coverage	False at delivery
Age guard <= 1.0 s	68.8% (322/468)	7.3% (34/468)
Prior global q99	2.1% (10/468)	0.0% (0/468)
New global q99	2.1% (10/468)	0.0% (0/468)
Conditional q99	1.9% (9/468)	0.0% (0/468)

Conditional q99 margins by increasing image-noise feature

Band 1: 1.81 cm from 3,638 training frames; Band 2: 2.62 cm from 3,638 training frames; Band 3: 2.71 cm from 3,638 training frames; The feature combines the detected pixel centers, local projection sensitivity, 2 px configured noise, and the minimum heuristic marker-quality score. The score is not a probability. Underpopulated bands use the pooled training q99.

Physical reaching at controller stop: 93/156. Conditional image-margin undercoverage: 50/15727 profile-specific delivered post-settle frame events; those events reuse camera captures across transport profiles. Neither metric is the policy's fresh visual-confirmation count.

READ THE EVIDENCE AND REPLAY THE PAIRED DEMO

```
uv run visual-servo-conditional-uncertainty-benchmark
```

```
uv run --extra demo python scripts/generate_conditional_uncertainty_demo.py
```

Image measurements choose the margin; simulation labels grade the decision.

All completion rules remain offline experiments. The fully delayed RGB rows delay target and tip together; they are not the current-joint-state hybrid.

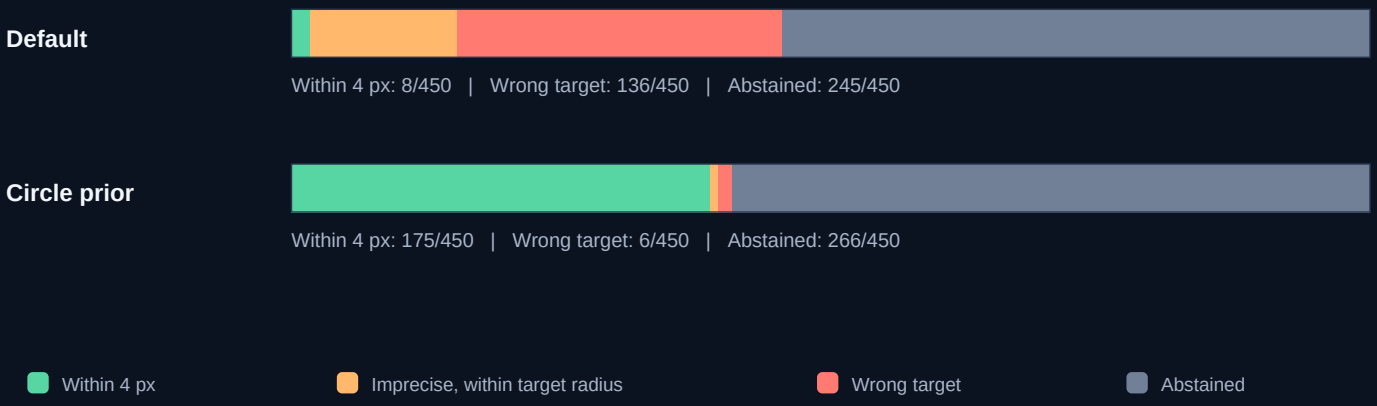
Maintainer-facing takeaway

A conditional margin is useful only if the held-out table shows a better coverage/false-confirmation trade-off without breaking the measured frame-level envelope. Keep the prior global q99 result visible even if the new calibration improves it.

A target-size prior helps, but identity still fails

An opt-in circle detector is compared with the unchanged largest-red-contour baseline on disjoint rendered lighting, distractor size and distance, and occlusion-width ranges.

PAIRED DETECTOR OUTCOMES ACROSS 450 HELD-OUT FRAMES



THE RED-DISK CONDITION REMAINS THE HARD CASE

Clutter	Default: strict / wrong / abstain	Circle prior: strict / wrong / abstain
none	8/150 | 0/150 | 84/150	76/150 | 0/150 | 72/150
red disk	0/150 | 66/150 | 82/150	10/150 | 6/150 | 134/150
red box	0/150 | 70/150 | 79/150	89/150 | 0/150 | 60/150

What improved, and what remains unresolved

The circle prior strictly localized 175/450 frames versus 8/450 for the default, and reduced wrong-target outputs from 136 to 6. It abstained on 266 frames. All six false accepts were small red disks under occlusion; in some frames the candidate's size-fit score was high even though it selected the distractor. The detector needs an identity cue such as a distinct marker, temporal tracking, or depth before it can guide control.

Trace the implementation

perception_candidate.py computes the 9 cm marker's projected radius, runs red-mask Hough circles, and abstains on competing circles. perception_candidate_validation.py randomizes scene factors and compares both detectors on identical frames. The default detector is unchanged; the experiment does not run the arm.

REPLAY THE VALIDATION

```
uv run --locked visual-servo-perception-candidate-validation --base-seed 9400
```

450 localization trials only | physical reaching and fresh visual confirmation not measured

Connecting perception to physical contact

The tri-arm extension reuses the project's detector and camera geometry, then evaluates a real MuJoCo contact task with the same free-payload scene as the scripted baseline.

01 RENDER

Top-down MuJoCo RGB; a visual-only red marker sits above the free payload.

02 DETECT + PROJECT

Unchanged detect_red_target() and pixel_to_table_xy() produce target XY.

03 PAIRED INPUT

Camera estimate plans the pick. Oracle XY uses ground truth only as a diagnostic.

04 PLAN THREE ARMS

Base yaw + elbow-up 3R IK + gripper yaw reuse the shared pick phases.

05 CONTACT + SCORE

Free payload, gravity, friction, and arm colliders drive lift and physical checks.

FOLLOW THE SHARED CODE PATH

run.py — detect_red_target

Existing OpenCV HSV detector returns the red marker center.

controller.py — pixel_to_table_xy

Existing pinhole projection maps that pixel to tabletop XY.

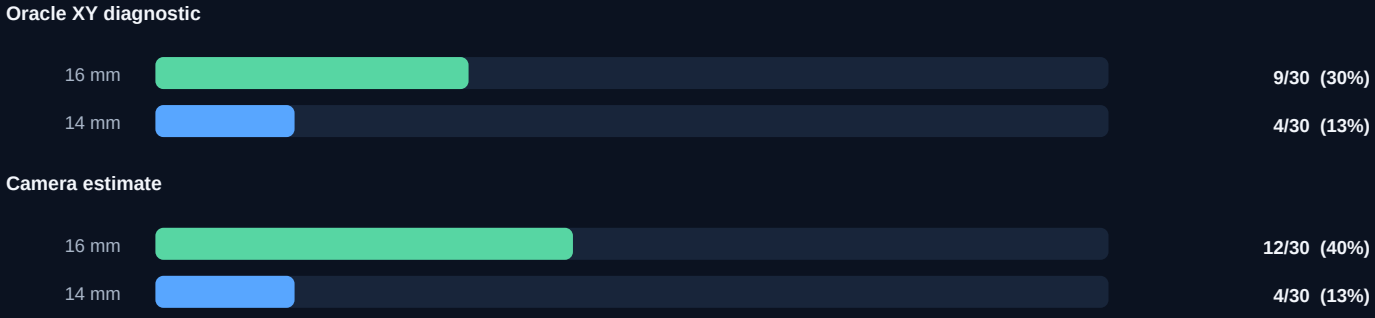
camera_guided_cooperative_pick.py — run_trial / _set_camera_guided_arm_controls

Pairs camera/oracle input and updates the three arm references.

cooperative_pick_demo.py — _arm_collision_masks / desired_pick_state

Reuses cross-arm colliders and task phases; visual wrist adapters, palm rails, and finger shanks connect to pads, which remain the grasp surfaces.

SAME-POSE HELD-OUT CONTACT-GEOMETRY COMPARISON



What the result points to next

On the same 30 poses, 16 mm pads lifted in 12/30 camera trials versus 4/30 at 14 mm; detector median error stayed 0.48 mm. Yet arm-environment and nonfinger payload contacts remain common. Improve collision-aware approach and grasp reliability before changing perception. This is one-shot camera-guided planning: physical lifting is not fresh visual confirmation, and the camera is not used during the lift.

A safer release follows the measured failure

Contact attribution showed that many successful holds became strict failures during release and withdrawal. The clearance profile changes only those actuator targets, preserving the free payload and active colliders.

1 IDENTIFY

Wrist-to-payload and fingertip-to-table contacts cluster during release and withdrawal.

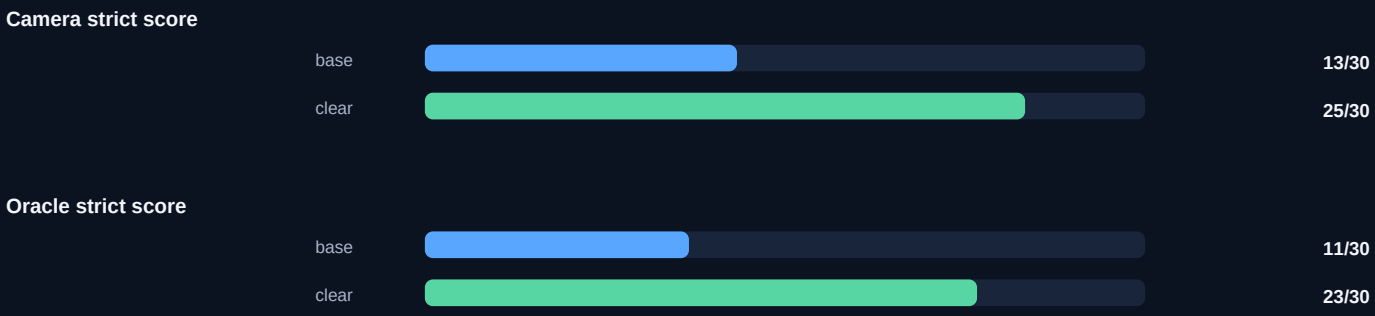
2 CLEAR

Raise the lowest open tool offset to 35 mm and keep tool centers at least 110 mm from the target.

3 REPLAY

Gravity, friction, arm colliders, and the strict full-cycle rule remain active.

FRESH PAIRED VALIDATION | SEED 38200 | 30 POSES



Payload-hold counts did not change: camera 26/30, oracle 23/30. The strict-score gains were paired: 12 camera poses and 12 oracle poses passed only under clearance; none passed only under baseline.

UNEXPECTED CONTACTS BY TRIAL

What changed in the simulator

Camera estimate: arm-table 7 -> 2; nonfinger arm-payload 14 -> 4. Oracle XY: arm-table 6 -> 0; nonfinger arm-payload 18 -> 5. Counts include any contact over the full trial. Remaining failures cluster in earlier grasp, lift, and lowering phases.

Interpretation and next work

The 35 mm and 110 mm settings were tuned after reviewing seed 38100; seed 38200 is a separate evaluation set, still only 30 poses. The selected demo pose is already in that aggregate. This measures physical contact and lift only: no new image confirms the payload after release. Broaden the evaluation and address remaining approach/lift contacts before claiming a generally reliable grasp.

`Code: _apply_release_profile() and run_trial() | Evidence: results/cooperative-pick-release-heldout-comparison.json`

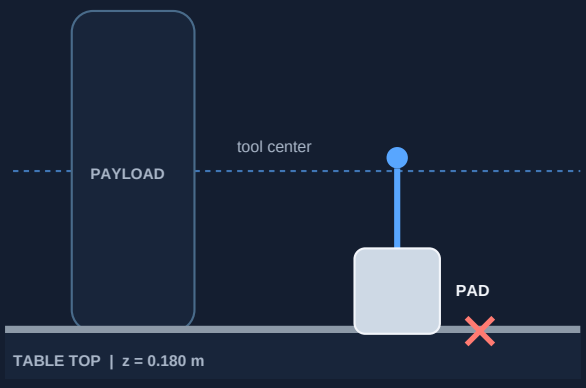
A 15 mm lift clears the lowest open pad

Contact traces revealed a table strike during the approach. Raising only that arm's open approach target removes the modeled overlap while leaving the finger geometry and closed grasp target unchanged.

READ THE CROSS-SECTION

5 mm baseline

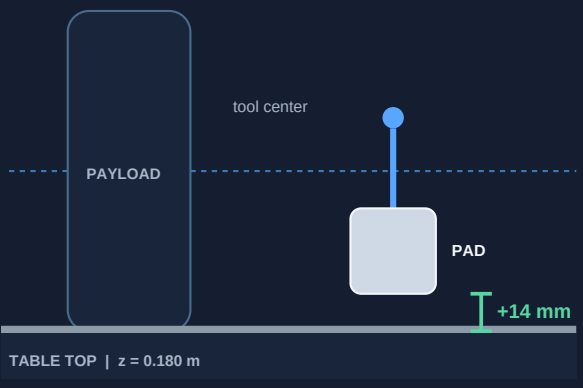
tool center = payload center + 5 mm



1 mm NOMINAL TABLE OVERLAP

20 mm candidate

tool center = payload center + 20 mm



14 mm TABLE CLEARANCE

EXACT PAIRED HELD-OUT SCORE | SEED 38300

Camera estimate

5 mm		27/30
20 mm		30/30

Wilson 95%: 74%-97% / 89%-100%

Paired: 3 candidate-only, 0 baseline-only

Oracle XY diagnostic

5 mm		28/30
20 mm		27/30

Wilson 95%: 79%-98% / 74%-97%

Paired: 2 candidate-only, 3 baseline-only

TRACE THE RESULT AND KEEP THE LIMITS IN VIEW

The paired failure

Pose 38301-003: arm0_right_pad contacts the table during lower-to-grasp and finger closure. The 5 mm run makes no stable grasp; the 20 mm replay passes the same physical score.

What changed

Camera score: 27/30 to 30/30, with three candidate-only wins and no baseline-only wins. Any camera-trial arm-table/ground contact falls from 2/30 to 0/30.

What remains uncertain

Oracle score moves 28/30 to 27/30. Camera confidence intervals overlap; n=30, ideal fixed calibration, no hardware, and no fresh post-pick visual confirmation.

`Code: _tool_height_offsets() / run_trial() | Evidence: approach-clearance-comparison.json and paired.csv`

The gain repeats across 60 new poses

A second seed block improves both camera-guided and oracle diagnostic scores. The paired reversals and overlapping confidence intervals still argue for caution.

Camera estimate

5 mm	52/60
20 mm	56/60

Wilson 95%: 76%-93% vs. 84%-97%

Paired wins/losses: 8 candidate-only / 4 baseline-only

Net change: +4 of 60

Oracle XY diagnostic

5 mm	53/60
20 mm	58/60

Wilson 95%: 78%-94% vs. 89%-99%

Paired wins/losses: 5 candidate-only / 0 baseline-only

Net change: +5 of 60

CONTACT TRIALS OVER THE COMPLETE PICK CYCLE

Camera estimate

Arm-table/ground: 4/60 → 2/60

Nonfinger arm-payload: 6/60 → 3/60

Trials with any contact; categories can overlap.

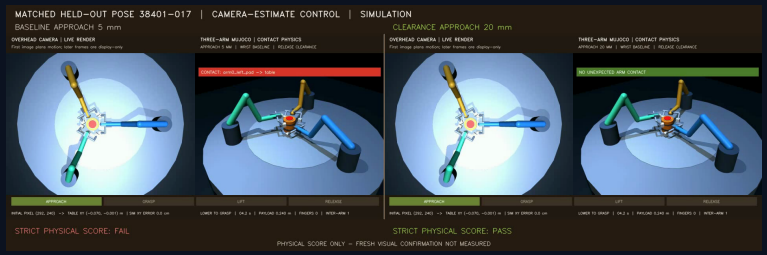
Oracle XY diagnostic

Arm-table/ground: 3/60 → 1/60

Nonfinger arm-payload: 6/60 → 2/60

Trials with any contact; categories can overlap.

MATCHED REPLAY AND INTERPRETATION



Pose 38401-017 | same camera-estimate target, fail → pass

Evidence boundary

This is a 60-pose MuJoCo split, not hardware evidence. The 4 camera baseline-only outcomes remain visible in the paired counts. Camera intrinsics are ideal and fixed; the target is detected once; fresh post-pick visual confirmation is not measured. Keep the 20 mm setting opt-in.

`Code: compare_cooperative_pick_geometry.py | Evidence: extended-comparison.json, extended-paired.csv, and paired replay metadata`

A second view checks the cooperative lift

After the initial overhead image plans the motion, a fixed side camera takes two fresh images. The image decision and the simulator's physical audit remain separate.

THREE RGB OBSERVATIONS FROM ONE SIMULATED PICK

START | 0.0 s



magenta fiducial | image-space measurement

FRESH | 8.4 s



magenta fiducial | image-space measurement

FRESH | 8.8 s



magenta fiducial | image-space measurement

CLASSIFIER CHANGES ON THE HELD-OUT SPLIT

Same frames for V0 / V1 / V2 / V3

Image rule	False confirm	Recall	Coverage
V0 rise + drift	0	100%	100%
V1 + fiducial angle	0	100%	100%
V2 + higher rise gate	0	100%	100%
V3 tighter angle gate	0	100%	100%

V0 checks rise and drift; V1 adds angle; V2 raises minimum rise to 50 px; V3 tightens angle to 4 degrees. All four match on seed 39400 (112 TP, 8 TN, zero errors or abstentions), so V3 shows no measured gain here. Eight negative labels leave specificity uncertain.

PHYSICAL OUTCOMES - SEPARATE SCORE

120 trials across two matched approach profiles

5 mm baseline	hold 53/60	strict 52/60
20 mm experimental	hold 59/60	strict 59/60

Same-pose strict comparison: 20 mm had 8 candidate-only wins and 1 baseline-only wins. Classifier labels, full hold, and full-cycle collision safety remain separate.

Evidence boundary and next failure

The first translation-only validation produced two false confirmations when the payload was highly tilted and had no grasp contact. Development then exposed an under-lift false confirmation, which motivated the 50 px rise gate. A later split found a 5.23 degree payload tilt that passed the 12 degree rule, so the angle gate was tightened to 4 degrees. Seed 39400 is the independent evaluation of V3. This remains an ideal synthetic camera with a visual-only fiducial and two samples; it does not prove continuous holding, safe release, or real-camera robustness. Keep the 20 mm approach opt-in.

`Code: cooperative_pick_visual_verification.py / cooperative_pick_visual_validation.py | Data: seed-39400 summary, CSV, and SVG scorecard`

Stress the camera path without changing the pick

V3 thresholds stay frozen while controlled faults alter only the side-camera RGB. The initial overhead control image and MuJoCo physics remain unchanged.

KEEP THE IMAGE AND PHYSICAL PATHS SEPARATE

CONTROL INPUT
Initial overhead RGB

→

OBSERVATIONS
Side RGB at 0.0 / 8.4 / 8.8 s

→

FAULT INJECTION
Post-capture image transforms

→

EVALUATION
MuJoCo truth and full motion

FROZEN V3 AGAINST CAPTURE-TIME LABELS | 58 POSITIVE / 2 NEGATIVE PER CONDITION

Image condition	False confirm	False reject	Abstain	Coverage	Recall	Specificity
Clean RGB	0	0	0	100%	100%	100%
Dim exposure x0.25	0	0	60	0%	0%	0%
Bright exposure x1.5	0	0	0	100%	100%	100%
50% marker occlusion	0	32	0	100%	45%	100%
Horizontal blur 9 px	0	1	0	100%	98%	100%
Fresh camera shift up 25 px	0	0	0	100%	100%	100%

Evidence boundary and next engineering step

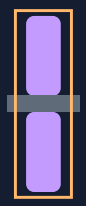
Seed 39600 covers 60 matched poses in 360 runs. The seed-39500 smoke check was excluded from validation. Dimming produced 60 abstentions and 0 false confirmations; a 25 px fresh-frame shift produced 0 false confirmations. Only two capture-time negatives were present, so observed specificity is weak evidence. Physical truth and full-motion outcomes matched across each pose's six repeats: yes. These are authored image transforms, not real-camera data. Next, test gap-tolerant marker components and an image-quality abstention gate on development seeds, then keep the change opt-in until a separate frozen split validates it.

Code: visual_image_faults.py / cooperative_pick_visual_fault_validation.py | Data: seed-39600 summary, CSV, and SVG scorecard

Rejoin a split fiducial with bounded evidence

The measured failure was partial occlusion: the marker split into components and V3 rejected real lifts. V4 groups aligned fragments while measuring only pixels that remain visible.

OBSERVED COMPONENTS

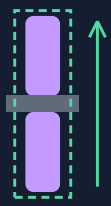


Middle band hidden

V3 chooses one

largest component

V4 BOUNDED UNION



28 px max gap

2 visible parts

union from pixels

A gap is not filled: the detector uses the coordinates of observed pixels for its center and PCA angle.

DEVELOPMENT | SEED 39700

30 matched poses | same V3 thresholds

Clean: V3 TP 28/29, FP 0, FN 1

V4 TP 28/29, FP 0, FN 1

50% occlusion: V3 TP 15/29, FP 0, FN 14

V4 TP 29/29, FP 0, FN 0

Physics unchanged by image condition: hold 29/30; strict 29/30

FROZEN VALIDATION | SEED 39800

60 matched poses | same V3 thresholds

Clean: V3 TP 55/56, FP 0, FN 1

V4 TP 55/56, FP 0, FN 1

50% occlusion: V3 TP 23/56, FP 0, FN 33

V4 TP 55/56, FP 0, FN 1

Physics unchanged by image condition: hold 55/60; strict 53/60

What the comparison supports

On the development split, the frozen 28 px option preserved all clean V3 decisions and raised occluded positive recall from 15/29 to 29/29. The separate seed-39800 split tests this one frozen option. Each split contains few negative labels; these results support better tolerance to this authored marker band only. V3 remains the default, and a visual confirmation remains separate from physical hold and full-cycle collision safety.

Code: cooperative_pick_visual_verification.py / cooperative_pick_occlusion_validation.py | Data: seeds 39700 and 39800, CSV and SVG scorecards

Gap tolerance helps only some occlusions

The 28 px option was frozen before this challenge. Each physical pose was simulated once, then nine image variants were evaluated from the same initial and two fresh side-camera frames.

NOMINAL PADS | 100 POSES

Capture-time truth: +94 / –6

Payload hold 94/100 | strict full cycle 92/100

14 MM HARD-NEGATIVE PADS | 100 POSES

Capture-time truth: +18 / –82

Payload hold 16/100 | strict full cycle 16/100

AUTHORED SIDE-CAMERA MASKS | SHOWN RELATIVE TO THE FIDUCIAL

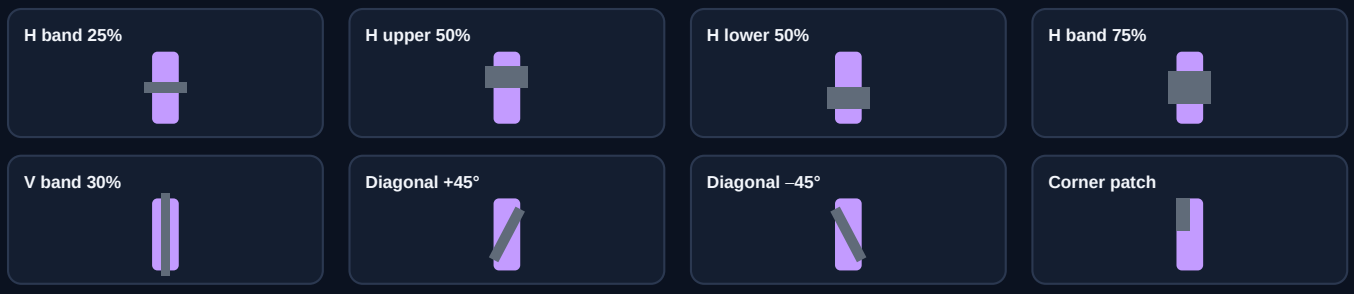


Image condition	16 mm positives V3 → V4 / 94	14 mm positives V3 → V4 / 18	False confirms V3 → V4 / 82 negatives
Clean	94 → 94	16 → 16	1 → 1
Horizontal 25% centered	1 → 94	1 → 16	0 → 1
Horizontal 50% upper	15 → 43	5 → 11	1 → 1
Horizontal 50% lower	27 → 62	3 → 13	0 → 0
Horizontal 75% centered	37 → 6	6 → 4	6 → 5
Vertical 30% centered	94 → 94	18 → 18	1 → 1
Diagonal +45°	8 → 8	6 → 6	2 → 2
Diagonal –45°	18 → 18	4 → 4	1 → 1
Upper-left patch	93 → 93	18 → 18	1 → 1

What the challenge changed

Moderate horizontal bands let V4 recover some true lifts, but the vertical, diagonal, and patch masks show no gain. At 75% band height, nominal true positives fell from 37/94 to 6/94. In the 14 mm hard-negative stratum, the 25% mask added one false confirmation and the severe mask still produced five among 82 capture-time negatives. This mask/pad mix is an authored stress test, not a deployment prevalence estimate. Keep V4 opt-in and V3 as the default. Capture-time visual truth remains separate from payload hold and full-cycle safety.

Code: visual_image_faults.py / cooperative_pick_occlusion_challenge.py | Data: seeds 40100 and 40200, 1,800 image rows

A second view catches a hidden tilt

The latest held-out experiment renders an actual MuJoCo rail across one camera's sight line, then checks the same free payload from two perpendicular side views.



Counted held-out replay: original camera rail, independent view, image verdict, and separate simulator audit

240	physical pose runs
121	capture-time negatives
0 / 121	two-view false confirms

Image rule	TP / 119	False confirms / 121	TN / 121	Abstain	Coverage
V3 | side	19	3	109	9	231/240
V4 | side 28px	101	4	108	9	231/240
V4 + orthogonal	100	0	95	26	214/240

16 MM NOMINAL PADS

Capture truth +98 / −2

Two-view TP 85 | false confirms 0 | abstain 2

14 MM HARD-NEGATIVE CONTROL

Capture truth +21 / −119

Two-view TP 15 | false confirms 0 | abstain 24

A safer decision, with more abstentions

The two-view option confirms 100/119 positives and makes 0/121 false confirmations (Wilson 95% upper bound 3.1%). It abstains on 26 poses and misses 19 positives; side-only V4 misses 18. Across the same 240 trajectories, 112/240 pass the full payload-hold check and 108/240 pass the strict full cycle. These physical outcomes are independent of the image decision. Keep V3 as the runtime default; V4 and two-view V4 remain opt-in.

Code: cooperative_pick_visual_verification.py / cooperative_pick_orthogonal_validation.py | Data: seeds 40500 and 40600, 240 poses

Contact has a friction threshold

A paired MuJoCo stress map separates the camera target estimate from the shared grasp and contact trajectory.

216 paired physics trials
base seed 40800 | same poses in every cell

12 shared target poses
base seed 40800 | same poses in every cell

9 mass/friction settings
base seed 40800 | same poses in every cell

CAMERA-ESTIMATED TARGET

pad/payload sliding friction	0.5	1.1	2.2
0.45 kg	0/12 H 0 / C 4	2/12 H 2 / C 3	11/12 H 11 / C 11
0.72 kg*	0/12 H 0 / C 8	0/12 H 0 / C 5	10/12 H 10 / C 10
0.99 kg	0/12 H 0 / C 8	0/12 H 0 / C 1	11/12 H 11 / C 12

Strict / 12 H = stable hold C = full-cycle collision-safe

ORACLE TARGET UPPER BOUND

pad/payload sliding friction	0.5	1.1	2.2
0.45 kg	0/12 H 0 / C 6	1/12 H 2 / C 3	12/12 H 12 / C 12
0.72 kg*	0/12 H 0 / C 6	0/12 H 0 / C 1	12/12 H 12 / C 12
0.99 kg	0/12 H 0 / C 8	0/12 H 0 / C 4	12/12 H 12 / C 12

Strict / 12 H = stable hold C = full-cycle collision-safe

* Nominal mass: 0.720116653454 kg; original compiled inertia preserved.

What the result says

At friction 0.5, neither controller achieved a strict success in any mass cell; all 72 runs were labeled no grasp contact. At friction 2.2, camera-estimated success was 11/12, 10/12, and 11/12 across light, nominal, and heavy payloads. Matching oracle failures at low friction point to contact mechanics or the shared trajectory, while the small, author-selected simulator sweep cannot establish real-object reliability.

`Visual confirmation: not measured | Code: cooperative_pick_material_validation.py | Data: 216 paired rows`

The CSV preserves strict lift, hold-only, and full-cycle collision outcomes as separate columns.

A larger pad did not restore hold contact

New-seed paired MuJoCo test: vary pad half-size while keeping nominal payload, friction, pose, approach, release, and controller matched.

96 paired physics trials
seed 40900 | camera estimate + oracle diagnostic

12 shared target poses
seed 40900 | camera estimate + oracle diagnostic

2 × 2 pad and friction settings
seed 40900 | camera estimate + oracle diagnostic

SLIDING FRICTION M = 0.5

Camera estimate

16 mm	0/12	hold 0 | full-cycle safe 6
18 mm	0/12	hold 0 | full-cycle safe 5

Oracle XY diagnostic

16 mm	0/12	hold 0 | full-cycle safe 5
18 mm	0/12	hold 0 | full-cycle safe 3

SLIDING FRICTION M = 2.2

Camera estimate

16 mm	12/12	hold 12 | full-cycle safe 12
18 mm	12/12	hold 12 | full-cycle safe 12

Oracle XY diagnostic

16 mm	12/12	hold 12 | full-cycle safe 12
18 mm	12/12	hold 12 | full-cycle safe 12

Measured conclusion

At friction 0.5, both pad sizes produced 0/12 strict lifts for camera and oracle; the legacy no_grasp_contact label means zero contacts in the scored hold window. At friction 2.2, both pad sizes scored 12/12 in both controllers. The 18 mm candidate did not improve low-friction hold contact, so the runtime 16 mm geometry stays in place.

NEXT: phase traces find contact in closure/early lift, then none in loaded hold; measure relative slip next.

Hold and full-cycle collision safety remain separate metrics; visual confirmation: not measured.

`Code: cooperative_pick_pad_friction_validation.py | Data: seed 40900, 96 paired rows`

The four-panel video replays one counted pose and adds no benchmark trials.

Contact disappears before hold

A new 48-trial block samples six named finger pads at each MuJoCo physics step to separate closure, cooperative lift, and loaded hold.

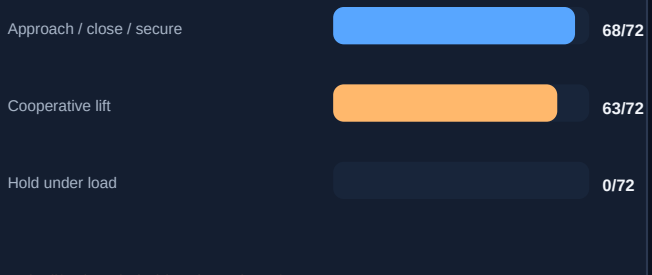
48 paired physics trials
seed 41000 | same poses at mu 0.5 and 2.2

288 per-finger traces
seed 41000 | same poses at mu 0.5 and 2.2

2 ms physics timestep
seed 41000 | same poses at mu 0.5 and 2.2

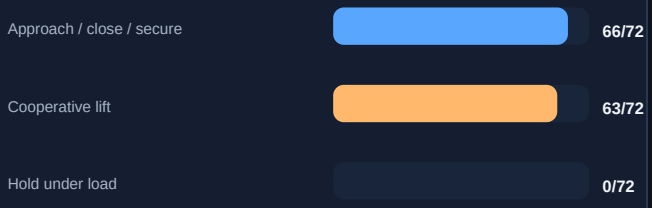
SLIDING FRICTION M = 0.5

Camera estimate



Strict lift 0/12 | hold 0 | cycle-safe 5

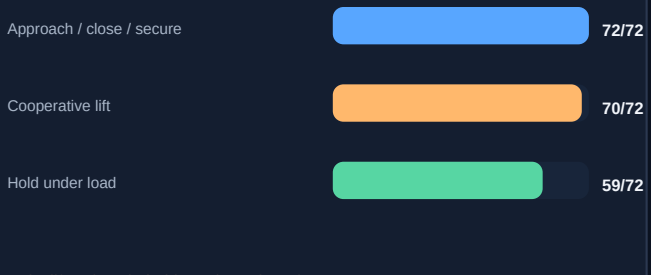
Oracle XY diagnostic



Strict lift 0/12 | hold 0 | cycle-safe 5

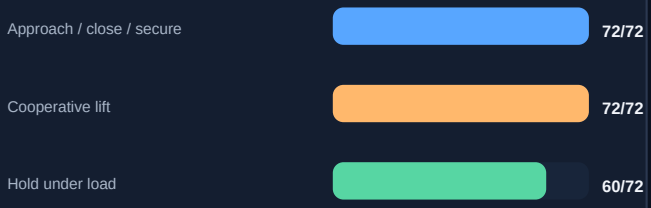
SLIDING FRICTION M = 2.2

Camera estimate



Strict lift 12/12 | hold 12 | cycle-safe 12

Oracle XY diagnostic



Strict lift 11/12 | hold 11 | cycle-safe 11

What the phase data resolves

At mu 0.5, both controllers show pad contact during closure and early lift, but none of the 24 trials retains any pad contact in the loaded hold. At mu 2.2, camera-estimate succeeds 12/12 and oracle succeeds 11/12. The low-friction failure is contact retention through lift; this study does not distinguish tangential slip from finger separation.

The old no_grasp_contact label means zero contacts in its scored hold window; it does not mean the fingers never touched.

Contact wrench is a MuJoCo solver output, not a calibrated sensor. Visual confirmation: not measured.

`Code: cooperative_pick_contact_diagnostics.py | Data: seed 41000, 48 paired rows`

Next: test one opt-in grip-preload change on new held-out poses.

The payload slips relative to closed fingers

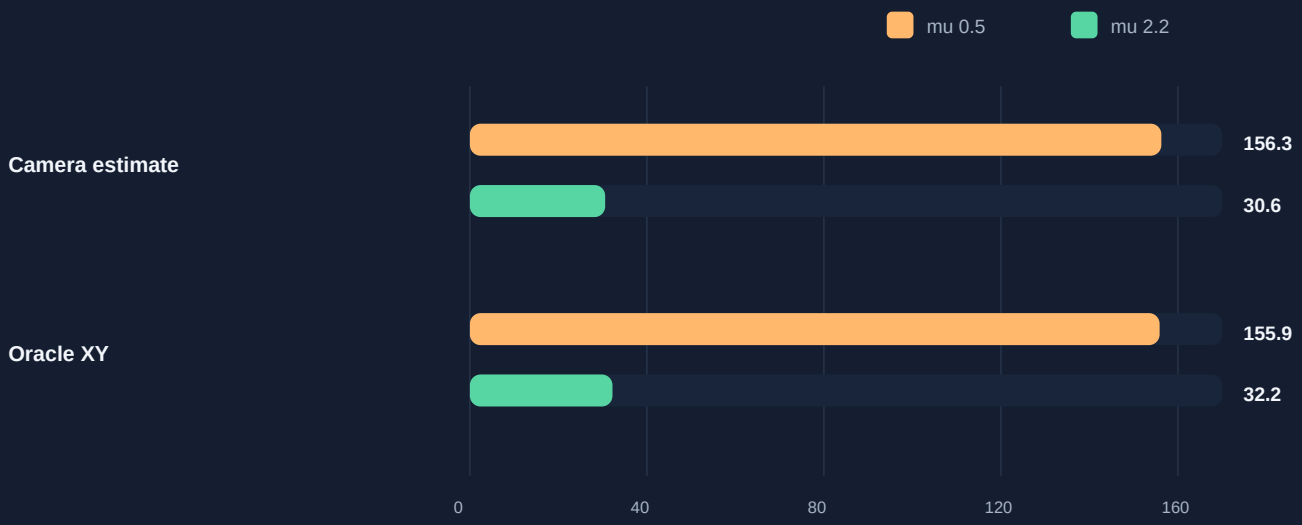
A fresh 48-trial split measures hand-relative payload motion, pad contact forces, and finger position tracking through the loaded hold.

48 paired physics trials
seed 42000 | 12 matched poses per condition

576 pad and phase records
seed 42000 | 12 matched poses per condition

20 ms diagnostic sample period
seed 42000 | 12 matched poses per condition

MEDIAN MAXIMUM DOWNWARD PAYLOAD DRIFT RELATIVE TO THE GRIPPER HAND (MM)



Evidence points to payload slip, not an open gripper

At mu 0.5, all 24 camera/oracle trials lost pad contact during hold. Median maximum finger-slide error during hold stayed at 0.019 mm (camera) and 0.020 mm (oracle). Contacting-sample tangential/normal force ratios were 0.468 and 0.470, near the selected friction value. At mu 2.2, both controllers lifted 12/12, with 31 and 32 mm median hand-relative drift.

Safety and confirmation remain separate

Only 6/12 camera and 7/12 oracle low-friction runs passed the full-cycle collision checks; at high friction 12/12 camera and 12/12 oracle runs passed. These are simulation outcomes under one scripted grasp and author-selected friction. No fresh visual confirmation was measured. Next, test a single opt-in grip-preload candidate on new matched poses and keep physical lift, collision safety, and image confirmation as separate scores.

Code: cooperative_pick_retention_diagnostics.py | Data: seed 42000, 48 trials, 576 pad-phase rows